Prob 7. (ZAT, Ch 7, 2.5)

(i) Using the results of the previous pr-m 6, show that a fundamental sol-n set (ODE terminology) for $R(r)$ (the radial component of the sol-n in spherical coordinates) for $r \neq 0$ is

(1) $$\{ r^{\alpha_1}, \; r^{\alpha_2} \}$$

where α_1, α_2 are the roots of the eq-n

(2) $$\alpha(\alpha - 1) - \mu = 0.$$

(ii) What is the fundamental sol-n set for $R(r)$ when $\mu = 0$?

Sol-n:

(i) By the result of Pr. 6, we have

(3) $$\frac{d}{dr}\left[r^2 \cdot R'(r)\right] - \mu R = 0 \quad \Rightarrow$$

$$2r \cdot R'(r) + r^2 R''(r) - \mu R = 0. \quad \Rightarrow \text{ reordering}$$

(4) $$r^2 \cdot R'' + 2 \cdot r \cdot R' - \mu R = 0.$$

(4) is 2nd order linear ODE, so let's assume the L.I. sol-ns for (4) is in the form of

(5) $$R(r) = r^{\alpha} \quad \text{where} \quad \begin{matrix} r \neq 0 \\ \alpha = \text{const.} \end{matrix}$$

[$r \neq 0$, in spherical $r \neq 0$ & if $r = 0$ $R(r) = 0$, $u = 0$, trivial as in Pr. 6]

then, R' & R'' are

(6) $$\begin{cases} R' = \alpha \, r^{\alpha - 1} \\ R'' = \alpha \cdot (\alpha - 1) \cdot r^{\alpha - 2} \end{cases} \quad \text{(using power rule)}$$

substituting (6) into (4) yields

(7) $$r^2 \cdot \alpha(\alpha - 1) \cdot r^{\alpha - 2} + 2 \cdot r \cdot \alpha \cdot r^{\alpha - 1} - \mu r^{\alpha} = 0 \quad \Rightarrow$$

$$\alpha \cdot (\alpha - 1) \cdot r^{\alpha} + 2\alpha \cdot r^{\alpha} - \mu \cdot r^{\alpha} = 0 \quad \Big| : r^{\alpha} \text{ BS} \qquad r \neq 0 \Rightarrow r^{\alpha} \neq 0$$

$$\Rightarrow \alpha(\alpha-1) + 2\alpha - \mu = 0$$

$$\alpha^2 - \alpha + 2\alpha - \mu = 0$$

$$\alpha^2 - \alpha - \mu = 0 \quad \Rightarrow$$

(8) $$\boxed{\alpha(\alpha-1) - \mu = 0}$$

$\alpha_i =$ roots of (8). So the fundamental sol-n set

$$\boxed{F.S.S = \{ r^{\alpha_1}, r^{\alpha_2} \}}$$ as desired. $\blacksquare$

(ii) When $\mu = 0$, (8) becomes

$$\alpha(\alpha-1) - 0 = 0$$

$$\alpha_1 = 0, \quad \alpha_2 = 1.$$

Then,

$$F.S.S = \{ r^0, r^1 \} = \{1, r\}.$$

By superposition principle the sol-n for (3) is then

$$\boxed{R(r) = C_1 \cdot 1 + C_2 \cdot r.}$$ $\blacksquare$